A multiscale scheme for the single-cluster bound

Working note

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8th November 2026

Abstract

We set up a multiscale renormalisation, in the spirit of Sznitman's and of Gracar and Stauffer's schemes as presented in Stauffer's lecture notes, for the single-cluster bound of the first-moment programme: the probability that a box of side L contains a thick crossing should be $O(\mu^{j_0} L^{-(j_0-2)})$. The scheme differs from the standard one in three respects forced by the model: the target is polynomial, so a box is declared bad when it contains a well-separated bad sub-boxes, with a a large constant and geometric scales $L \mapsto L^\gamma$; the dependence between far sub-boxes is through lines rather than through mixing particles, and is handled by conditioning on the lines shared between the sub-boxes, which are few and automatically aligned; and the induction hypothesis carries a finite set of fixed lines, organised into aligned groups, with a weight $1 + K^s \mu^{-s} L^{s-2}$ for a group of s lines, which is the amount by which s fixed aligned lines can help a crossing. We prove the recursion under this hypothesis, with explicit constraints on the parameters, for configurations of tight groups and singletons in general position, and we give proofs, complete up to bookkeeping, of the two single-scale statements it needs, for nearly complete groups and for the base scale. The scheme is not complete: the class of fixed-line configurations on which the hypothesis is asserted must be closed under the recursion, and we exhibit partially aligned configurations that are inherited by sub-boxes but on which the bound is false, so that the class and its weights have to be refined; we describe the refinement we expect to work. Unlike every first-moment argument over covers, the scheme never counts sub-configurations of a cluster, so the decoration obstruction does not arise. The dependence step from the single-cluster bound to the thick-cluster lemma remains open.

1 Setting and result

$\mathcal{L}$ is a Poisson line process with intensity measure $\nu = (\mu/\pi)\,\mathrm{d}\theta\,\mathrm{d}p$, $N(z)$ the number of lines meeting $Q(z) = z + [0,1)^2$, $\mu_1 = 4\mu/\pi$, and a box is thick at level j if at least j lines meet it. For a finite set G of lines (*fixed lines*) we write $N_G(z)$ for the number of lines of $\mathcal{L} \cup G$ meeting $Q(z)$ and $\Theta_G(j)$ for the set of boxes with $N_G(z) \geq j$. A *box of side* L is $B = z + [-L/2, L/2)^2$ with $z \in \mathbb{Z}^2$, its *inner box* is $z + [-L/4, L/4)^2$, and a *crossing* of B at level j is a 2-connected set of boxes of $\Theta_G(j)$ that meets the inner box and a box at sup-distance at least $L/2 - 1$ from z. Let

$$\mathrm{Bad}(B, G) := \{B \text{ contains a crossing at level } j_0 \text{ in } \mathcal{L} \cup G\}\,.$$

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If the thick cluster of 0 has diameter at least d, then the box of side d centred at 0 satisfies $\mathrm{Bad}(B,\varnothing)$. The single-cluster bound of [HJ26b] therefore follows from

$$\mathbb{P}(\mathrm{Bad}(B,\varnothing)) \le C\,\mu^{j_0} L^{-(j_0-2)} \qquad \text{for all boxes of side } L \ge 1\,. \tag{1.1}$$

The right-hand side is the probability that j_0 lines of $\mathcal{L}$ run within distance 1 of each other across B: there are of order L^2 such bundle positions and each costs $(\mu/L)^{j_0}$; it is the true order.

Aligned groups. A set g of lines is an *aligned group at scale L in B* if all its lines lie within distance 1 of one line (its *axis*) throughout B. A finite set G of lines meeting B is *organised at scale L* if it is partitioned into aligned groups $g_1,\dots,g_m$, lines belonging to no group counting as groups of size one; we write $s_i = |g_i|$ and

$$W_L(G) \coloneqq \prod_{i=1}^{m} w_L(s_i), \qquad w_L(s) \coloneqq 1 + K^s \mu^{-s} L^{s-2}\,, \tag{1.2}$$

with a constant $K \ge 1$ fixed below. The weight is what fixed lines are worth: s fixed aligned lines reduce the cost of a bundle along their axis from $(\mu/L)^{j_0}L^2$ to $(\mu/L)^{j_0-s}$, a factor $\mu^{-s}L^{s-2}$. A single fixed line is worth only $1 + K/(\mu L)$, because the bundle it could complete has lost its freedom of direction.

Definition 1.1 (Induction hypothesis)**.** $\mathcal{H}(L)$ holds if for every box B of side L and every set G of at most $L^{1/2}$ fixed lines organised at scale L in B,

$$\mathbb{P}(\mathrm{Bad}(B,G)) \le C_0\,\mu^{j_0} L^{-(j_0-2)}\, W_L(G)\,. \tag{1.3}$$

Theorem 1.2 (Multiscale recursion)**.** *Let $\gamma = 3$ and $j_0 \ge j_1$, let $a \ge a_1(j_0)$ and $K \ge K_1(j_0)$, and let $\mu \le \mu_1(j_0,a,K)$. There is $L_1 = L_1(j_0,a,K,\mu)$ such that the following holds. Let $\mathcal{G}$ be a class of fixed configurations that is inherited by sub-boxes in the sense of Section 6. If $\mathcal{H}(L')$ holds on $\mathcal{G}$ for all $L' \le L/8$ with a constant $C_0 \ge C_1(j_0,a,K)$, if $L \ge L_1$, and if the nearly-complete-group estimate of Lemma 5.1 holds at scale L, then $\mathcal{H}(L)$ holds on $\mathcal{G}$ with the same constant C_0. Together with the base case of Lemma 5.2, this gives (1.1) with $C = C_0$. The class of configurations consisting of tight groups and finitely many singletons satisfies all requirements except that it is not known to be inherited; see Section 6.*

The recursion is carried out in Section 4. Lemmas 5.1 and 5.2 are stated in Section 5 with proofs that we regard as complete up to the bookkeeping indicated there; they are single-scale statements and contain no induction. Section 6 discusses the class $\mathcal{G}$, which is where the scheme is not yet complete.

2 Crossings and separated sub-boxes

Fix L, $\ell \coloneqq \lceil L^{1/\gamma} \rceil$ and an integer $a \ge 2$. Cover B by the boxes of side ℓ centred at the points of $z + (\ell/2)\mathbb{Z}^2$ (the *sub-boxes*; each point of B lies in the inner box of one of them), and put $D \coloneqq L/(8a)$.

Lemma 2.1 (A crossing gives separated bad sub-boxes)**.** *On $\mathrm{Bad}(B,G)$ there are sub-boxes $B_1,\dots,B_a$, pairwise at distance at least D, such that $\mathrm{Bad}(B_i, G\cap B_i)$ holds for every i, where $G \cap B_i$ denotes the lines of G meeting B_i. The number of such a-tuples is at most $(4L/\ell)^{2a}$.*

Proof. Let S be a crossing of B. Since S is 2-connected, the sub-boxes whose inner boxes meet S form a chain $B'_1, B'_2, \dots$ in which consecutive members are at distance at most ℓ, from a sub-box meeting the inner box of B to one meeting its boundary layer, so the chain has length at least $L/(4\ell)$. For each member B' of the chain, S meets the inner box of B' and, since $\operatorname{diam} S \geq L/4 > \ell$, leaves B'; the component of $S \cap B'$ containing a box of the inner box is 2-connected and reaches the boundary layer of B', so it is a crossing of B' in $\mathcal{L} \cup G$, and only the lines of $\mathcal{L} \cup G$ meeting B' are relevant for it. Taking every $\lceil L/(4a\ell) \rceil$-th member of the chain gives a sub-boxes pairwise at distance at least $L/(4a) - 2\ell \geq D$ for $L \geq L_1$. The count is the number of ways to choose a sub-boxes among $(2L/\ell + 1)^2$. □

3 Shared and private lines

Let $B_1, \dots, B_a$ be sub-boxes pairwise at distance at least D. A line of $\mathcal{L}$ is *shared* if it meets at least two of them and *private to* B_i if it meets B_i and no other B_j. Let $\mathcal{S}$ be the set of shared lines and $\mathcal{P}_i$ the set of private lines of B_i.

Lemma 3.1 (Conditional independence)**.** *Given $\mathcal{S}$, the processes $\mathcal{P}_1, \dots, \mathcal{P}_a$ are independent Poisson processes, and*

$$\mathbb{P}\Big(\bigcap_{i \leq a} \mathrm{Bad}(B_i, G \cap B_i)\Big) \leq \mathbb{E}\Big[\prod_{i \leq a} \mathbb{P}\big(\mathrm{Bad}(B_i, (G \cap B_i) \cup (\mathcal{S} \cap B_i)) \mid \mathcal{S}\big)\Big], \tag{3.1}$$

where inside the conditional probability the lines of $\mathcal{S} \cap B_i$ are treated as fixed and the randomness is that of $\mathcal{P}_i$.

Proof. The sets of lines meeting exactly one, and at least two, of the B_i are disjoint, and a Poisson process restricted to disjoint sets gives independent processes. A crossing of B_i uses only lines meeting B_i, which are private, shared or fixed. □

Lemma 3.2 (Shared lines are few and aligned)**.** *Let B_i, B_j be sub-boxes at distance $D' \geq D$. The lines meeting both have ν-measure at most $4\mu\ell^2/D'$. Any two of them make an angle at most $2\ell/D'$. For a line ℓ_0 meeting B_i, the lines meeting B_j that lie within distance 1 of ℓ_0 throughout B_i have measure at most $8\mu\ell/D'$.*

Proof. A line meeting two sets of diameter $\sqrt{2}\ell$ at distance D' has direction within angle $\sqrt{2}\ell/D'$ of the direction joining their centres and, for each direction, offset in an interval of length $\sqrt{2}\ell$; the measure is at most $(\mu/\pi) \cdot 2\sqrt{2}\ell/D' \cdot \sqrt{2}\ell \leq 4\mu\ell^2/D'$. The angle bound is the same computation. For the last claim the offset interval has length 2 instead of $\sqrt{2}\ell$, and the direction must be within $\sqrt{2}\ell/D'$ of a fixed direction, giving $(\mu/\pi) \cdot 2\sqrt{2}\ell/D' \cdot 2 \leq 8\mu\ell/D'$. □

With $D = L/(8a) = \ell^\gamma/(8a)$ and $\gamma = 3$, the expected number of shared lines through a given sub-box is at most $a \cdot 4\mu\ell^2/D = 32a^2\mu/\ell$, and the measure of shared lines aligned with a given fixed line is at most $64a^2\mu/\ell^2$. Shared lines meeting a sub-box B_i and a far sub-box B_j have directions within $2\ell/D \leq 1/\ell$ of each other, so two of them that are within lateral distance 1 in B_i form an aligned group at scale ℓ in B_i automatically.

4 The recursion

Throughout, $\gamma = 3$, $\ell = L^{1/3}$, $D = L/(8a)$, and $\rho_L := \mu^{j_0} L^{-(j_0-2)}$, so that $\rho_\ell^a = \mu^{aj_0}\ell^{-a(j_0-2)}$ and $\rho_L = \mu^{j_0}\ell^{-3(j_0-2)}$. We first record the arithmetic that the scheme rests on, then the three ways in which sub-boxes can be bad.

The basic inequality. For the bound (1.3) to propagate from scale ℓ to scale L through Lemmas 2.1 and 3.1, we need

$$(4L/\ell)^{2a}\,(2C_0\rho_\ell)^a \le C_0\rho_L\cdot\ell^{-\delta} \quad \text{for some } \delta>0\,, \tag{4.1}$$

the factor $\ell^{-\delta}$ being the slack that absorbs constants and the error terms below. In terms of exponents of ℓ this is $2a(\gamma-1)-a(j_0-2)\le-\gamma(j_0-2)-\delta$, i.e.

$$a\big(j_0-2-2(\gamma-1)\big) \;\ge\; \gamma(j_0-2)+\delta\,, \tag{4.2}$$

which holds for $j_0>2\gamma=6$ and a large; the μ-powers satisfy $\mu^{aj_0}\le\mu^{j_0}$, and $(8C_0)^a4^{2a}$ is absorbed by $\ell^{-\delta/2}$ for $\ell\ge\ell_0(C_0,a)$. With two sub-boxes ($a=2$) and super-exponential scales the left side of (4.2) is negative: this is why the standard scheme cannot give a polynomial bound, and why we use a well-separated sub-boxes and geometric scales.

Rows. For a group g of G with axis A_g, its *row* R_g is the set of sub-boxes met by a line of g; these are within distance $\ell\sqrt{2}+1$ of A_g, and g is an aligned group at scale ℓ in each of them. A sub-box outside all rows carries no fixed line.

Case 1: a separated sub-boxes outside all rows. Each is bad with fixed set consisting of shared lines only. By Lemma 3.1 and $\mathcal{H}(\ell)$, the probability is at most $(4L/\ell)^{2a}\mathbb{E}\prod_i(C_0\rho_\ell W_\ell(\mathcal{S}\cap B_i))$, and by Lemma 4.1 below the expectation is at most $(2C_0\rho_\ell)^a$. By (4.1) this is at most $C_0\rho_L\ell^{-\delta}\le C_0\rho_L W_L(G)/4$.

Case 2: a separated sub-boxes on the row of one group g, $x:=j_0-|g|\ge3$. Each sub-box sees g as a group of size $s=|g|$ at scale ℓ, and by $\mathcal{H}(\ell)$ is bad with probability at most $C_0\rho_\ell(1+k_\ell(s))\cdot$(shared-line factor), where $k_\ell(s)=K^s\mu^{-s}\ell^{s-2}$ and $\rho_\ell k_\ell(s)=K^s\mu^x\ell^{-x}$. The sub-boxes lie on a row, so there are at most $(4L/\ell)^a$ choices, not $(4L/\ell)^{2a}$. The bound is

$$(4L/\ell)^a\big(4C_0K^s\mu^x\ell^{-x}\big)^a=(16C_0K^s)^a\mu^{xa}\ell^{a(\gamma-1)-ax}\,,$$

to be compared with the term $C_0\rho_Lk_L(s)=C_0K^s\mu^xL^{-x}=C_0K^s\mu^x\ell^{-\gamma x}$ of the target. The ratio is at most $(16C_0K^s)^{a-1}\mu^{x(a-1)}\ell^{\gamma x-a(x-\gamma+1)}$, whose exponent is negative iff

$$a\;\ge\;\frac{\gamma x}{x-\gamma+1}=\frac{3x}{x-2}\,, \tag{4.3}$$

which for $x\ge3$ holds with $a\ge9$, with slack $\ell^{-\delta'}$ absorbing the constants for $\ell\ge\ell_0$. For $x\le2$ the exponent is non-negative and the recursion does not close; these are the nearly complete groups of Lemma 5.1.

Case 3: mixed. The a sub-boxes may lie on several rows, and some on none. Order the groups so that the sub-boxes on the row of g_1 are the most numerous; since W_L is additive over groups, the target contains the term $C_0\rho_Lk_L(|g_1|)$, and the product over sub-boxes is bounded by the product of their individual bounds, each of which is at most $C_0\rho_\ell(1+\sum_g k_\ell(|g|))\le C_0\rho_\ell(m+1)\max_g k_\ell(|g|)$. Writing x^* for the smallest x among the groups whose rows contain chosen sub-boxes, each sub-box costs at most $(m+1)C_0K^{j_0}\mu^{x^*}\ell^{-x^*}$, and the combinatorial factor is at most $(4L/\ell)^{2a}$ for the sub-boxes off rows and $(4L/\ell)^a$ along rows; the worst case is covered by (4.3) with x^* and the slack. The factor $(m+1)^a$, with m the number of groups, is the one place where the number of fixed groups enters; it is absorbed if $m\le\ell^{\delta/(4a)}$, see the discussion in Section 6.

Lemma 4.1 (Shared lines change the weights little)**.** *Let B_i be one of the separated sub-boxes with inherited fixed set $G_i = G \cap B_i$, organised at scale ℓ by the groups of G. Organise $G_i \cup (\mathcal{S} \cap B_i)$ at scale ℓ by adding each shared line that lies within distance 1 of the axis of a group of G_i throughout B_i to that group, grouping shared lines that lie within distance 1 of a common line throughout B_i, and leaving the remaining shared lines as singletons. Then*

$$\mathbb{E}\big[W_\ell(G_i \cup (\mathcal{S} \cap B_i))^2\big]^{1/2} \le 2\, W_\ell(G_i) \qquad (\ell \ge \ell_0(a, K, \mu))\,.$$

Proof. By Lemma 3.2 the shared lines through B_i form a Poisson process of mean at most $\lambda = 32a^2\mu/\ell$, and those within distance 1 of a given line throughout B_i have mean at most $\lambda' = 64a^2\mu/\ell^2$. A group of G_i of size s gains s' shared lines with probability at most $\lambda'^{s'}/s'!$, and $k_\ell(s+s') = k_\ell(s)(K\ell/\mu)^{s'}$, so the second moment of its weight is at most $k_\ell(s)^2 \sum_{s'} (\lambda' K\ell/\mu)^{2s'}/s'! \le k_\ell(s)^2 \exp\{(64a^2K/\ell)^2\}$. A new group of $s' \ge 2$ shared lines around a common axis has, for each of at most ℓ axis positions at unit resolution and a single direction class, probability at most $(\lambda/\ell)^{s'}$, and weight $k_\ell(s') = K^{s'}\mu^{-s'}\ell^{s'-2}$; the expected sum of squared weights is at most $\ell \sum_{s'\ge 2} (\lambda/\ell)^{s'} k_\ell(s')^2 = \ell^{-3} \sum_{s'\ge 2} (32a^2K^2\mu^{-1})^{s'} \le \ell^{-3} C(a, K, \mu)$. Singleton shared lines contribute $1 + K/(\mu\ell)$ each, with second moment at most $\exp\{3\lambda K/(\mu\ell)\} = \exp\{96a^2K/\ell^2\}$. Collecting the three contributions and using $W_\ell \ge 1$ gives the claim for $\ell \ge \ell_0$. □

The lemma is stated for the second moment because the weight of a new group of s' shared lines grows like $\ell^{s'-2}$ while its probability decays like $\ell^{-2s'}$, so that higher moments are not uniformly controlled. In Case 1 the product over the a sub-boxes is accordingly bounded by pairing the factors and applying Cauchy–Schwarz $\lceil \log_2 a \rceil$ times, which replaces the constant 2 by $2a$ and is absorbed by the slack.

5 The two single-scale statements

Lemma 5.1 (Nearly complete groups)**.** *Let G be organised at scale L and let $g \in G$ have $x = j_0 - |g| \le 2$. Assume $\mathcal{H}(L')$ for all $L' \le L/16$. Then*

$$\mathbb{P}(\mathrm{Bad}(B, G)) \le \mathbb{P}(E_g) + C_2 K^{|g|} \mu^x L^{-x}\,,$$

where E_g is the event that a crossing exists that visits fewer than $L/32$ boxes met by all lines of g, and the first term is bounded by Cases 1–3 of Section 4 applied to the sub-boxes that the crossing visits outside the row of g, of which there are at least $L/(8\ell)$.

Proof sketch. Call a box *on* g if it is met by all lines of g; on-boxes lie within distance 2 of the axis A_g and need x further lines of $\mathcal{L} \cup (G \setminus g)$ to be thick. If a crossing S has fewer than $L/32$ on-boxes, then since S has at least $L/8$ boxes, at least $3L/32$ of its boxes are off g, and its chain of sub-boxes has at least $L/(8\ell)$ members outside the row of g, which are bad with the fixed set $G \setminus g$; this is the first term. If S has at least $L/32$ on-boxes, each of them is met by x lines of $\mathcal{L} \cup (G \setminus g)$. A line meeting two on-boxes at distance t lies within distance 2 of A_g between them, so each line covers an interval of on-boxes, and the on-boxes are covered x-fold by these intervals. For a random line the interval has length t with ν-measure at most $C\mu L/t^2$ summed over positions, so the expected number of random lines with interval length in $[t, 2t]$ is at most $C\mu L/t^2$; an x-fold cover of a set of at least $L/32$ on-boxes by intervals of length at most $2t$ needs at least $xL/(64t)$ of them, of probability at most $(C'\mu t/L)^{xL/(64t)}$ by the Poisson tail, which summed over dyadic $t \le L/128$ is $O((\mu/L)^{2x})$, while $t \ge L/128$ gives the x lines of length comparable to L within distance 2 of A_g, of probability at most $(C\mu/L)^x$ with the constant absorbing the at most 128

positions. The fixed lines of $G \setminus g$ cover on-boxes as well: a line not in g lies within distance 2 of A_g along a stretch of length at most $4/\sin\alpha$, α its angle with A_g, and if G is organised so that lines outside g make an angle at least $L^{-1/2}$ with A_g, these stretches have length at most $4L^{1/2}$, so the fixed lines cover at most $4L^{1/2}|G|$ on-boxes, which is less than $L/64$ when $|G| \le L^{1/2}/256$; the random lines must then cover at least $L/64$ on-boxes x-fold, and the argument applies with $L/32$ replaced by $L/64$. □

Lemma 5.2 (Base case)**.** *There are $L_1 = L_1(j_0)$ and $\mu_1 = \mu_1(L_1)$ such that $\mathcal{H}(L)$ holds for all $L \le L_1$ and $\mu \le \mu_1$ with a constant $C_0 = C_0(j_0)$ that does not depend on L_1.*

Proof sketch. By [HJ26a, Proposition 6.2], applied to the crossing event in place of the diameter event, $\mathbb{P}(\mathrm{Bad}(B, G))$ is at most the expected number of minimal sets $\Lambda \subseteq \mathcal{L}$ such that $\Lambda \cup G$ produces a crossing, which is $\sum_n M_n/n!$ with M_n a $\nu^{\otimes n}$-integral over n-tuples of lines meeting B. Each line meeting B has measure at most $4\mu L$, so $M_n/n! \le (4\mu L)^n/n!$, and the terms with $n > 2j_0$ sum to at most $(4e\mu L/(2j_0))^{2j_0}$, which is at most ρ_L for $\mu \le cL^{-5}$. For $n \le 2j_0$ the thick set of $\Lambda \cup G$ is the union over subsets of j_0 lines of their common boxes, and the common boxes of a subset lie within the 1-neighbourhood of any one of its lines, a straight tube. A crossing has diameter at least $L/4$ and is covered by at most $r := \binom{2j_0+|G|}{j_0}$ such tubes, so one tube covers a part of it of diameter at least $L/(4r)$: j_0 lines of $\Lambda \cup G$ pass within distance 1 of two boxes at distance $t \ge L/(4r)$. If s of them are fixed and belong to one group g, the remaining $j_0 - s$ random lines pass within distance 1 of the axis of g along a stretch of length t at one of at most $4r$ positions, of measure $(C\mu/t)^{j_0-s} \le C' r^{j_0} (\mu/L)^{j_0-s}$, which is $C'' \rho_L k_L(s) L^{-2}$, below the target; if none is fixed, the first random line has measure at most $4\mu L$ and the others $(C\mu/t)^{j_0-1}$, giving $C' r^{j_0} \mu^{j_0} L^{2-j_0} = C' r^{j_0} \rho_L$. The remaining $n - j_0$ lines of Λ contribute at most $(4\mu L)^{n-j_0}$, and the sum over $n \le 2j_0$ is at most $2C' r^{j_0} \rho_L W_L(G)$ for $4\mu L \le 1/2$. Fixed lines from different groups passing within distance 1 of two boxes at distance t are excluded by the organisation of G when $t \ge L^{1/2}$ and $L \ge L_1$. This gives $C_0 = 2C' r^{j_0}$, which depends on j_0 and on $|G|$ but not on L. □

The base case shows where the L-dependence of the bound comes from: a crossing of a box of constant side L_1 with few lines forces j_0 lines to run together over a length comparable to L_1, and that costs $(\mu/L_1)^{j_0-1}$ per bundle. A bound of the form $C(L_1)\mu^{j_0}$ with the right power of μ but an uncontrolled constant would not do: the first recursion step multiplies constants by the power a, and only the slack $\ell^{-\delta}$ of (4.1), which is a power of L_1, can absorb them.

6 The class of fixed configurations, and what is not proved

Everything above is conditional on a correct formulation of the class of fixed-line sets G for which $\mathcal{H}(L)$ is asserted. The class must satisfy three requirements: it must be *inherited*, in that $G \cap B_i$ together with the shared lines is in the class at scale ℓ, up to events of probability at most ρ_L; it must be *closed* under the organisation of Lemma 4.1; and $\mathcal{H}(L)$ must be *true* on it. We record what we know.

1. *Groups.* A tight group, lines within distance 1 of a common axis throughout B, is inherited as a tight group in every sub-box of its row. Shared lines join a group at cost $\mu\ell^2/D$ against a gain $K\ell/\mu$, product $\ell^2/D = 8a/\ell$ (Lemma 4.1). The additive weight (1.2) is the right form: a crossing uses one group over most of its length, and two groups in sequence cost the product of two bundle probabilities, which is far below the sum of the two weights.

2. *Singletons.* A fixed line belonging to no group is worth $1 + K/(\mu L)$ at most if it is in *general position*, and in the recursion its row is handled by Case 2 with $x = j_0 - 1$, where the multiplicative factor plays no role. The hypothesis must therefore allow any finite number of singletons, and a cardinality bound $|G| \le L^{1/2}$ is not inherited: the lines of G meeting a sub-box number up to $|G|$, while the sub-box's own allowance is only $\ell^{1/2} = L^{1/6}$. A density condition, $|G \cap B'| \le L'^{1/2}$ for every box $B' \subseteq B$ of side $L' \ge L_1$, *is* inherited, and the shared lines respect it except on an event of probability at most $\ell^2 (C\mu/\ell)^{L_1^{1/2}} \le \rho_L$ for $L_1 \ge (4j_0)^2$.

3. *Partial alignment, and a counterexample.* Let g be a group of $j_0 - 1$ lines and let G contain in addition $L^{1/2}$ single lines, each within distance 1 of the axis of g along a stretch of length $L^{1/2} - 1$, the stretches tiling the axis. Then every box on g is met by j_0 fixed lines, $\mathrm{Bad}(B, G)$ holds with probability one, and G satisfies the density condition. Any definition of "aligned" with a threshold stretch length $\tau(L)$ admits this counterexample with tiles of length $\tau(L) - 1$, provided $L/\tau(L)$ tiles are allowed by the density condition; and the threshold cannot be a constant, because then a shared line joins a group at cost $\mu\ell^2/(t_0 D)$ against the gain $K\ell/\mu$, product $8aK/t_0$, which is not small.

4. *What is needed.* A class of configurations closed under the recursion in which such tilings are either forbidden or correctly weighted. The natural candidate is to weight *every* set of fixed lines by the amount it can shorten a crossing: for a line ℓ_0 and a group g, the length along the axis of g over which ℓ_0 lies within distance 1 of it, and more generally for each stretch I of a potential axis the number of fixed lines within distance 1 along I; the weight of G would then be a sum over axes and stretches of $k_{|I|}$(number of lines) times the cost of a bundle completing the crossing outside I. We expect this to be the right object, and that with it the three requirements can be met, but we have not carried out the verification, and we do not claim Theorem 1.2 beyond the class of configurations consisting of tight groups and singletons in general position with respect to all axes at all scales, for which the counterexample does not arise but which is not known to be inherited.

Why the scheme is nevertheless the right one. The recursion never counts sub-configurations of a cluster: the entropy at each scale is the fixed factor $(4L/\ell)^{2a}$, and the probability of a bad box is controlled by the induction, not by a first moment. The decoration obstruction of [HJ26a], which rules out every Peierls argument over covers for $d \gg 1/p_0$, therefore does not arise, and the bundle structure of the model, which defeated the product bounds of [HJ26c], appears here only in the weights of fixed lines, where it is a one-dimensional bookkeeping along an axis rather than a two-dimensional one along a cluster. The remaining difficulty, the formulation of the class of fixed configurations, is a single, explicit, combinatorial problem.

7 The multiplicity weight

The counterexample of Section 6 is removed, and most of the requirements on the class are met, by replacing the group weights by a weight that measures what the fixed lines can do along any one of them.

Definition 7.1 (Multiplicity weight)**.** For a finite set G of lines meeting B, a line $\ell_0 \in G$ and a point p of $\ell_0 \cap B$, let $n_G(\ell_0, p)$ be the number of lines of G, including ℓ_0, that pass within distance 3

of p, and let $m_G(\ell_0) \coloneqq \max_p n_G(\ell_0, p)$. The *multiplicity weight* is

$$W_L(G) \coloneqq 1 + \sum_{\ell_0 \in G} k_L(m_G(\ell_0)), \qquad k_L(m) \coloneqq K^m \mu^{-m} L^{m-2}. \tag{7.1}$$

A box within distance 1 of ℓ_0 at p is met by at most $n_G(\ell_0, p) \le m_G(\ell_0)$ fixed lines, so it needs at least $j_0 - m_G(\ell_0)$ random lines to be thick; this is the only property of the weight that the arguments use. For a tight group of s lines, $m = s$ for each of its lines and the weight is $s\,k_L(s)$, as before up to the factor s. For the tiling configuration of Section 6, $m = j_0$ along the axis, $k_L(j_0)\rho_L = K^{j_0} \ge 1$, and the bound is true. For fixed lines in general position, $m \le 2$ and the weight is bounded. The weight is inherited: for a sub-box B_i, $m_{G\cap B_i}(\ell_0) \le m_G(\ell_0)$, since the maximum runs over fewer points and fewer lines, and the shared lines through B_i raise the multiplicity of a fixed line by the number of them that pass within distance 3 of it at a common point, whose cost against the gain $K\ell/\mu$ is the ℓ^2/D of Lemma 4.1. Note that no assumption on G remains: $\mathcal{H}(L)$ is asserted for every finite G.

Lemma 7.2 (Many fixed lines through a sub-box force multiplicity)**.** *Let G' be a set of lines meeting a box of side ℓ, and let $M \coloneqq \max_{\ell' \in G'} m_{G'}(\ell')$. Then $|G'| \le \ell M$, and consequently $\sum_{\ell' \in G'} k_\ell(m_{G'}(\ell')) \le \ell M k_\ell(M) \le k_\ell(M+1)$ for $K \ge \mu M$.*

Proof. The lines of G' cross one of the two diagonals of the box, say in at least $|G'|/2$ points on a segment of length $\ell\sqrt{2}$; some sub-segment of length 6 contains at least $|G'| \cdot 6/(2\ell\sqrt{2}) \ge 2|G'|/\ell$ of them, and the corresponding lines are within distance 3 of a common point on one of them, so $M \ge 2|G'|/\ell$. The second claim follows from $k_\ell(M+1) = k_\ell(M) K\ell/\mu$. □

Light and heavy sub-boxes. Call a sub-box B_i of the chain of Lemma 2.1 *light* if $M_i \coloneqq \max_{\ell' \in G\cap B_i} m_{G\cap B_i}(\ell') \le j_0 - 6$ (or $G \cap B_i = \varnothing$) and *heavy* otherwise. By Lemma 7.2 and $\mathcal{H}(\ell)$, a light sub-box is bad with probability at most $C_0\rho_\ell(1 + k_\ell(j_0 - 5)) \cdot$ (shared-line factor) $\le 4C_0 K^{j_0-5}\mu^5\ell^{-5}$.

Case L: at least $L/(8\ell)$ light sub-boxes in the chain. Choose a of them pairwise at distance at least $D = L/(8a)$; by Lemma 3.1 and the above, the probability is at most $(4L/\ell)^{2a}(4C_0K^{j_0-5}\mu^5\ell^{-5})^a = C^a\ell^{4a-5a} = C^a\ell^{-a}$, which is at most $C_0\rho_L$ once $a \ge 3(j_0 - 2) + \delta$ and $\ell \ge \ell_0$. The weights of the heavy lines are not used at all.

Case H: at least $L/(8\ell)$ heavy sub-boxes in the chain. A heavy sub-box lies within distance $3 + \ell\sqrt{2}$ of a point where at least $j_0 - 5$ fixed lines pass within distance 3 of a common point on a fixed line ℓ'; call such ℓ' *heavy*, and let $R(\ell')$ be the set of sub-boxes met by ℓ'. The heavy sub-boxes of the chain are covered by the rows $R(\ell')$ of heavy lines. Consecutive chain sub-boxes in the same row form *runs*.

- If some run has at least $a\ell^{3/2}$ sub-boxes, choose a of them pairwise at distance at least $D' \coloneqq \ell^{5/2}$ along the row. Shared lines between two of them that pass within distance 3 of ℓ' have measure at most $C\mu\ell/D'$ by Lemma 3.2, against a gain of $K\ell/\mu$ in the weight, product $CK\ell^2/D' = CK\ell^{-1/2}$; so the shared-line factor is at most 2, and the bound is $(4L/\ell)^a(4C_0\rho_\ell k_\ell(m))^a$ with $m = m_G(\ell')$, which is Case 2 of Section 4 and closes for $x = j_0 - m \ge 3$ by (4.3); for $x \le 2$, Lemma 5.1 applies to ℓ'. The sum over the heavy line ℓ' matches the sum in (7.1).
- Otherwise all runs have fewer than $a\ell^{3/2}$ sub-boxes, so the chain changes rows at least $L/(8a\ell^{5/2}) = L^{1/6}/(8a)$ times. A change from $R(\ell')$ to $R(\ell'')$ either passes through a light sub-box, or happens within distance 2ℓ of the crossing point of ℓ' and ℓ''; in the first case, if

this happens a times, Case L applies. There remains the case of a crossing that hops at least $L^{1/6}/(8a) - a$ times between heavy lines at their crossing points, riding each for fewer than $a\ell^{5/2} = aL^{5/6}$ boxes.

The hopping case. The crossing follows a path $\ell'_1, \ell'_2, \dots, \ell'_q$ in the *crossing graph* of the heavy lines, switching from ℓ'_i to ℓ'_{i+1} at their crossing point, and on the stretch of ℓ'_i between two switches, of length t_i, every box within distance 1 of ℓ'_i needs at least $x_i = j_0 - m(\ell'_i) \geq 1$ random lines. The switch points are determined by G; at a switch point the number of heavy lines to continue on is at most $n_G(\ell'_i, p) \leq m(\ell'_i) \leq j_0$, since $m \geq j_0$ makes the bound trivial. Start the path at the point where the crossing leaves the inner box. The expected number of such configurations is, by the Mecke formula, at most

$$\sum_{\text{start}} \sum_{q \geq 1} j_0^q \sum_{(t_1, \dots, t_q)} \prod_{i=1}^{q} \Big(\frac{C\mu}{t_i}\Big)^{x_i},$$

where the inner sum runs over the realisable stretch lengths, with $\sum_i t_i \geq L/4$, and where lines serving two stretches are ignored (see below). Some stretch has $t_i \geq L/(4q)$; bounding it by $(4C\mu q/L)^{x_{\min}}$ and the others by $\sum_{t \geq 1} (C\mu/t)^{x_{\min}} = (C\mu)^{x_{\min}} \zeta(x_{\min})$ each gives

$$\sum_{\text{start}} \Big(\frac{4C\mu}{L}\Big)^{x_{\min}} \sum_{q} q^{1+x_{\min}} \big(j_0 (C\mu)^{x_{\min}} \zeta(x_{\min})\big)^{q-1} \leq C'' \sum_{\text{start}} \Big(\frac{\mu}{L}\Big)^{x_{\min}}$$

for $j_0 (C\mu)^{x_{\min}} \zeta(x_{\min}) \leq 1/2$, which holds for $\mu \leq \mu_1(j_0)$. The sum over the starting heavy line matches the sum in (7.1), and $(\mu/L)^{x_{\min}}$ is $\rho_L k_L(m)$ up to constants for the heaviest line used. No restriction on the number of heavy lines or on their crossings is needed: the entropy of the path is paid stretch by stretch by the factor $(C\mu)^x \zeta(x) < 1/(2j_0)$, and the length is paid once by the longest stretch. Two points remain in this computation. A random line may serve the end of one stretch and the beginning of the next, near their crossing point; this reduces the number of random lines by one per switch, by a factor at most $(t/(C\mu))^x$ for the shorter of the two stretches only if the two heavy lines make an angle below $1/t$, in which case they are within distance 3 of each other along the stretch and the multiplicity, hence x, already accounts for it; we have not written this out. And the hopping case was isolated under the assumption that the chain has fewer than $L/(8\ell)$ light sub-boxes, while a crossing may alternate between heavy stretches and light sub-boxes adjacent to them; the light sub-boxes are then not separated from the stretches, so the decoupling by shared lines does not apply to them, and their cost has to be charged differently, for instance by treating the random lines of the adjacent stretch as fixed lines for the light sub-box, which the multiplicity weight allows.

Status. With the multiplicity weight the induction hypothesis is asserted for all finite G, the weight is inherited and closed under shared lines (Lemma 4.1 applies with groups replaced by multiplicity classes), the tiling counterexample is absorbed, the recursion closes in Case L and in Case H with a long run, and the hopping case is bounded by a first moment over paths in the crossing graph that gives the right order without any restriction on the heavy lines. What is not complete is the bookkeeping at the interface: light sub-boxes adjacent to heavy stretches, and random lines shared between consecutive stretches. We regard both as technical, and we regard the multiplicity weight as the correct induction hypothesis for the scheme.

8 The two-parameter hypothesis and the proof architecture

The interface problems of Section 7 come from asking the hypothesis a single question, "is the box bad", when the recursion needs to distinguish crossings that use the fixed lines from crossings that avoid them. We make the hypothesis answer both.

Definition 8.1 (Avoiding crossings)**.** Let $\varepsilon_L := c_\varepsilon/(\log L)^3$ with $c_\varepsilon \le 1/2$ fixed. For fixed sets $G' \subseteq G$ of lines meeting B, let $\mathrm{Bad}(B; G, G')$ be the event that B contains a crossing 2-path P at level j_0 in $\mathcal{L} \cup G$, from the inner box to the boundary layer, such that fewer than $\varepsilon_L L$ boxes of P lie within distance 1 of a line of $G \setminus G'$.

Thus $\mathrm{Bad}(B; G, G) = \mathrm{Bad}(B, G)$, since every crossing contains a crossing 2-path, and $\mathrm{Bad}(B; G, \varnothing)$ is the event of a crossing that essentially ignores the fixed lines: a crossing path has at least $L/4$ boxes and $\varepsilon_L L \le L/4$.

Definition 8.2 (Two-parameter hypothesis)**.** $\mathcal{H}_2(L)$ holds if for every box B of side L and all finite $G' \subseteq G$ of lines meeting B,

$$\mathbb{P}\big(\mathrm{Bad}(B; G, G')\big) \le C_0\, \rho_L\, W_L(G')\,, \tag{8.1}$$

with the multiplicity weight (7.1) computed for G' inside B.

The content of the refinement is that lines of $G \setminus G'$ cost nothing: a crossing that is within distance 1 of them on a vanishing fraction of its boxes needs j_0 random lines on the rest, and the rest is a positive fraction of a crossing.

Lemma 8.3 (Avoidance is inherited)**.** *Let P be a crossing path of B with fewer than $\varepsilon_L L$ boxes within distance 1 of $G \setminus G'$. Then among the sub-boxes of its chain, at least $L/(8\ell)$ are such that the sub-path P_i, the part of P from its first box in the inner box of B_i to its first exit from B_i, has fewer than $\varepsilon_\ell \ell$ boxes within distance 1 of $G \setminus G'$.*

Proof. Each P_i is a crossing path of B_i with at least $\ell/4$ boxes, and a box of P lies in at most four sub-boxes, so $\sum_i |P_i \cap \mathcal{N}| \le 4|P \cap \mathcal{N}| < 4\varepsilon_L L$, where $\mathcal{N}$ is the set of boxes within distance 1 of $G \setminus G'$. If N_{bad} sub-paths have at least $\varepsilon_\ell \ell$ boxes in $\mathcal{N}$, then $N_{\mathrm{bad}} \varepsilon_\ell \ell < 4\varepsilon_L L$, i.e. $N_{\mathrm{bad}} < (4\varepsilon_L/\varepsilon_\ell)(L/\ell) = (4/27)(L/\ell)$, using $\varepsilon_\ell = \varepsilon_L(\log L/\log \ell)^3 = 27\varepsilon_L$. The chain has at least $L/(4\ell)$ members, and $L/(4\ell) - (4/27)(L/\ell) > L/(8\ell)$. □

The recursion for $\mathcal{H}_2$. Let S be a crossing realising $\mathrm{Bad}(B; G, G')$. By Lemma 8.3, at least $L/(8\ell)$ chain sub-boxes have sub-paths avoiding $G \setminus G'$ at level ε_ℓ. For each such sub-box B_i, let $G'_i := G' \cap B_i$ and let $\mathcal{S}_i$ be the shared lines through B_i. The sub-crossing either avoids G'_i as well, or uses it. In the first case $\mathrm{Bad}(B_i; G_i \cup \mathcal{S}_i, \mathcal{S}_i)$ holds, and $\mathcal{H}_2(\ell)$ gives the bound $C_0 \rho_\ell W_\ell(\mathcal{S}_i)$, whose expectation over the shared lines is at most $2C_0\rho_\ell$ by Lemma 4.1: the inherited fixed lines cost nothing, which is exactly what the one-parameter hypothesis could not provide. In the second case $\mathrm{Bad}(B_i; G_i \cup \mathcal{S}_i, G'_i \cup \mathcal{S}_i)$ holds, with the bound $C_0\rho_\ell W_\ell(G'_i \cup \mathcal{S}_i)$, and the sub-crossing has at least $\varepsilon_\ell \ell/4$ boxes within distance 1 of a line of G'_i: it rides a line of G' for a stretch. The classification of Section 7 now applies to G' in place of G:

1. If at least $L/(16\ell)$ of these sub-boxes are of the first kind, a separated ones give $(4L/\ell)^{2a}(2C_0\rho_\ell)^a \le C_0\rho_L \ell^{-\delta}$ by (4.1). This is Case L, now without any condition on the fixed lines through the sub-boxes.

2. Otherwise at least $L/(16\ell)$ sub-paths ride lines of G'. A ride along $\ell' \in G'$ inside B_i covers at least $\varepsilon_\ell \ell/4$ boxes within distance 1 of ℓ', each needing $j_0 - m_{G'}(\ell')$ random lines, and consecutive rides along the same line form runs. A run of at least $a\ell^{3/2}$ sub-boxes is Case 2 for ℓ' (or Lemma 5.1 when $x \le 2$); the entropy of the run's sub-boxes is one position each along the row, and the weight term $k_L(m_{G'}(\ell'))$ in (7.1) is the target. Otherwise the crossing changes the line it rides at least $L^{1/6}/(8a)$ times, and the hopping computation of Section 7 applies: the stretches of rides are joined either at crossing points of lines of G', with at most j_0 choices each, or by excursions of the crossing away from G'.

3. An excursion of length t between two rides is a connected set of t boxes none of which is within distance 1 of G', hence a crossing of a box of side $t/2$ avoiding G', with fixed lines the inherited $G \setminus G'$ that it also avoids, and the random lines of the adjacent rides, at most $2j_0$ of them, which may enter the excursion's box. For $t \ge L_1$, $\mathcal{H}_2(t/2)$ with the latter as G' bounds its probability by $C_0\rho_{t/2}W_{t/2}(\text{ride lines}) \le C_0\rho_{t/2}k_{t/2}(2j_0)$ when the ride lines form a single aligned group, and otherwise by less; this is at most $C_0K^{2j_0}\mu^{-j_0}t^{\,j_0-2}\cdot\mu^{j_0}(t/2)^{2-j_0} \le C_0(2K)^{2j_0}$, a constant, which is useless for short excursions but enters the hopping sum only as a bounded factor per switch, while the excursion's own length is a stretch with $x = j_0$ in the sum. The position of the excursion is determined by the rides it joins up to $O(1)$ choices. For $t < L_1$ the excursion is a constant-size set of boxes each needing j_0 random lines outside the ride lines, and a direct bound $(C\mu)^{j_0}$ per box applies after removing the at most $2j_0$ ride lines, which make a box thick only if $j_0 \le 2j_0$ of them pass through it, i.e. at the switch point itself.

Theorem 8.4 (Recursion for the two-parameter hypothesis)**.** *Let $\gamma = 3$, $j_0 \ge j_1$, $a \ge a_1(j_0)$, $K \ge K_1(j_0)$ and $\mu \le \mu_1(j_0, a, K)$. If $\mathcal{H}_2(L')$ holds for all $L' \le L/8$ with a constant $C_0 \ge C_1$, if $L \ge L_1$, and if the claims listed below hold at scale L, then $\mathcal{H}_2(L)$ holds with the same C_0. Together with the base case, this gives (1.1) with $\beta = j_0 - 2$.*

The claims. The proof of Theorem 8.4 is the case analysis above together with the computations of Sections 4 and 7. The following statements are used and are not proved in this note.

(C1) *Nearly complete lines.* For $\ell' \in G'$ with $m_{G'}(\ell') \ge j_0 - 2$, the direct coverage bound of Lemma 5.1 holds in the two-parameter setting: the probability that a crossing rides ℓ' on at least $L/32$ boxes is at most $C_2K^m\mu^x L^{-x}$ with $x = j_0 - m \le 2$, where the fixed lines of $G \setminus G'$ contribute to the coverage of at most $\varepsilon_L L \le L/64$ boxes by the avoidance assumption, so that the random lines must cover at least $L/64$ boxes x-fold.

(C2) *Shared service.* In the hopping sum, a random line within distance 1 of two consecutive rides near their switch point reduces the number of required random lines by one; this is compensated by the multiplicity of the two lines of G' near the switch point when their angle is below the reciprocal of the shorter stretch, and otherwise affects only a bounded number of boxes.

(C3) *Base case.* Lemma 5.2 in the two-parameter form: for $L \le L_1$ and $\mu \le \mu_1(L_1)$, $\mathbb{P}(\mathrm{Bad}(B; G, G')) \le C_0\rho_L W_L(G')$ with $C_0 = C_0(j_0)$; the longest-tube argument applies verbatim, the tube being along a line of $\mathcal{L} \cup G'$, since tubes along lines of $G \setminus G'$ are excluded by the avoidance assumption for the part of the crossing they would have to carry.

(C4) *Counting constants.* The constants in Lemma 8.3 and in the choice of separated sub-boxes along runs are compatible: a run of $a\ell^{3/2}$ sub-boxes contains a sub-boxes pairwise at distance $\ell^{5/2}$, and the shared lines between two of them that pass within distance 3 of the ridden line have measure at most $C\mu\ell/\ell^{5/2}$, as used in Section 7.

Assessment. (C1) and (C3) are the single-scale lemmas of Section 5 restated for the two-parameter hypothesis, and we see no new difficulty in them. (C4) is arithmetic. (C2) is the one claim that needs a genuine argument, and it is local: it concerns the neighbourhood of a switch point and two fixed lines. With the two-parameter hypothesis the interface problems of Section 7 are gone: a sub-box in which the crossing avoids the inherited fixed lines is charged ρ_ℓ times the weight of its shared lines only, excursions between rides are crossings of smaller boxes with the ride lines as fixed lines, and the counting of avoiding sub-boxes is inherited through the absolute threshold $\varepsilon_L L$ with $\varepsilon_L = c_\varepsilon/(\log L)^3$, for which $\varepsilon_\ell = 27\varepsilon_L$ gives the factor the counting needs. We regard Theorem 8.4 with (C1)–(C4) as the final form of the argument for the single-cluster bound. The dependence step from the single-cluster bound to the thick-cluster lemma, [HJ26b, Section 3.1], is not addressed by this note.

9 Summary

1. The multiscale scheme replaces the standard "two bad sub-boxes" recursion by "a well-separated bad sub-boxes" with geometric scales $L \mapsto L^3$, which is what a polynomial target requires ((4.2)), handles the dependence between far sub-boxes by conditioning on the shared lines, which are few and automatically aligned (Lemmas 3.1 and 3.2), and carries fixed lines in the induction hypothesis with additive weights (1.2). The recursion closes for groups missing at least three lines ((4.3)) and for sub-boxes outside all rows; nearly complete groups (Lemma 5.1) and the base case (Lemma 5.2) are single-scale statements with proofs we regard as complete up to the bookkeeping indicated.

2. The induction hypothesis with group weights is not correctly formulated: partially aligned singletons produce a counterexample for every scale-dependent alignment threshold (Section 6). The multiplicity weight of Section 7, which charges each fixed line by the maximal number of fixed lines near it, removes the counterexample, needs no restriction on G, is inherited, and closes the recursion in all cases but one: a crossing hopping between many heavy fixed lines at their crossing points with short runs on each. That case is bounded by a first moment over paths in the crossing graph of the heavy lines, which gives the right order with no restriction on G. The interface between rides along fixed lines and the rest of a crossing is handled by the two-parameter hypothesis of Section 8, which distinguishes the fixed lines a crossing uses from those it avoids and charges only the former; avoidance is inherited through the choice $\varepsilon_L = c_\varepsilon/(\log L)^3$. The final form of the argument is Theorem 8.4, which rests on the four claims (C1)–(C4) listed there, of which only (C2), a local statement about random lines serving two consecutive rides near a switch point, requires a new argument.

3. Two further points remain as before: the technical restriction to second moments in Lemma 4.1, and the dependence step from the single-cluster bound to the thick-cluster lemma in [HJ26b].

Acknowledgements

BJ acknowledges the financial support of the Leibniz Association within the Leibniz Junior Research Group on Probabilistic Methods for Dynamic Communication Networks (DYCOMNET) as part of the Leibniz Competition (grant no. J105/2020), of the Berlin Mathematics Research Center MATH+ through the project EF-MA-Sys-2 on Information Flow & Emergent Behavior in Complex Networks, and of the Deutsche Forschungsgemeinschaft (DFG) through Project P27 within the SPP 2265.

Use of AI tools

The simulations, proofs, computations, figures and the text of this note were developed with substantial assistance of the AI systems Claude (Anthropic) and ChatGPT Astra 6 (OpenAI), in an interactive process directed by the authors.

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